

17. *Assertion (A) :* When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

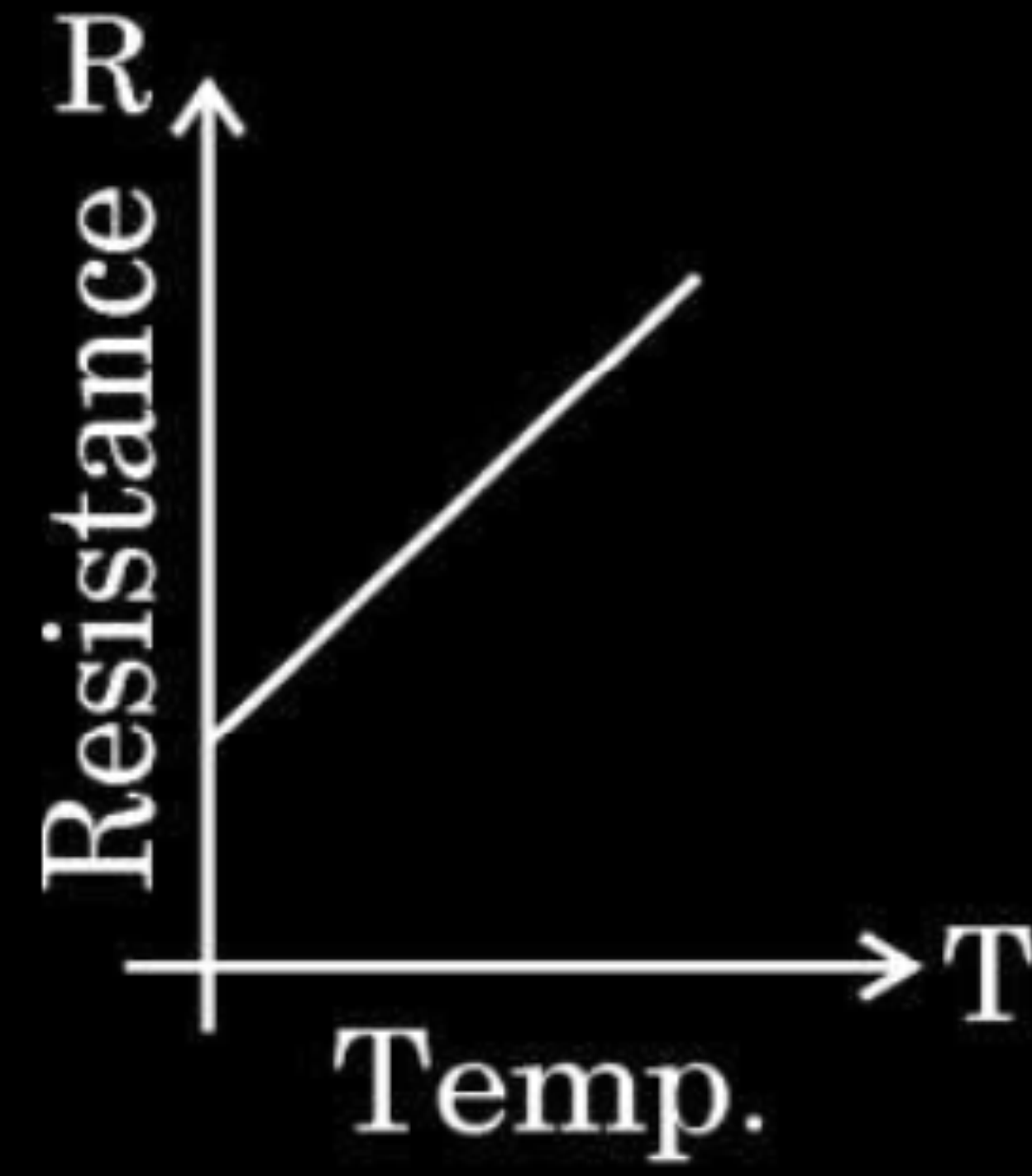
Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

17. *Assertion (A) :* When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

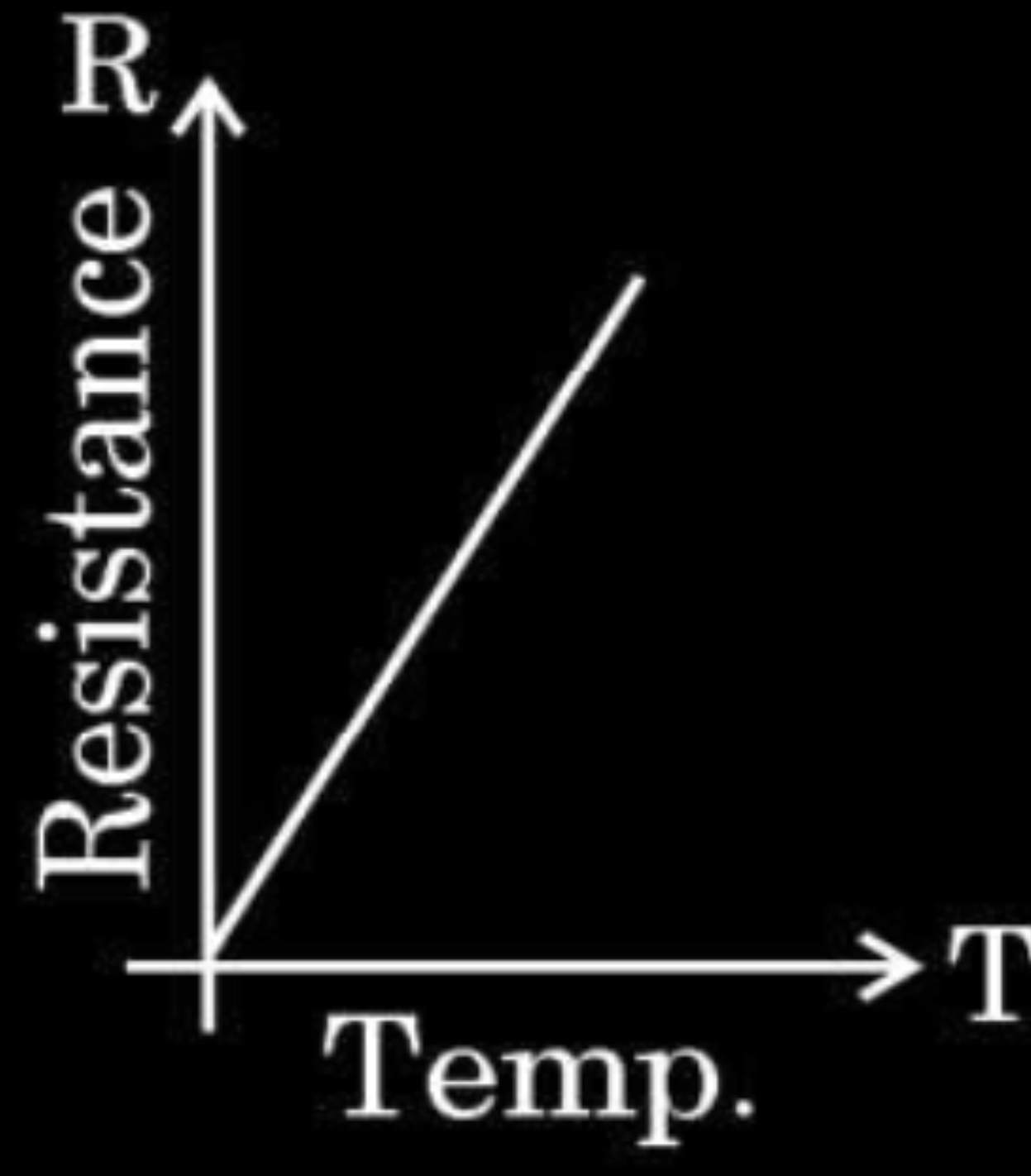
Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

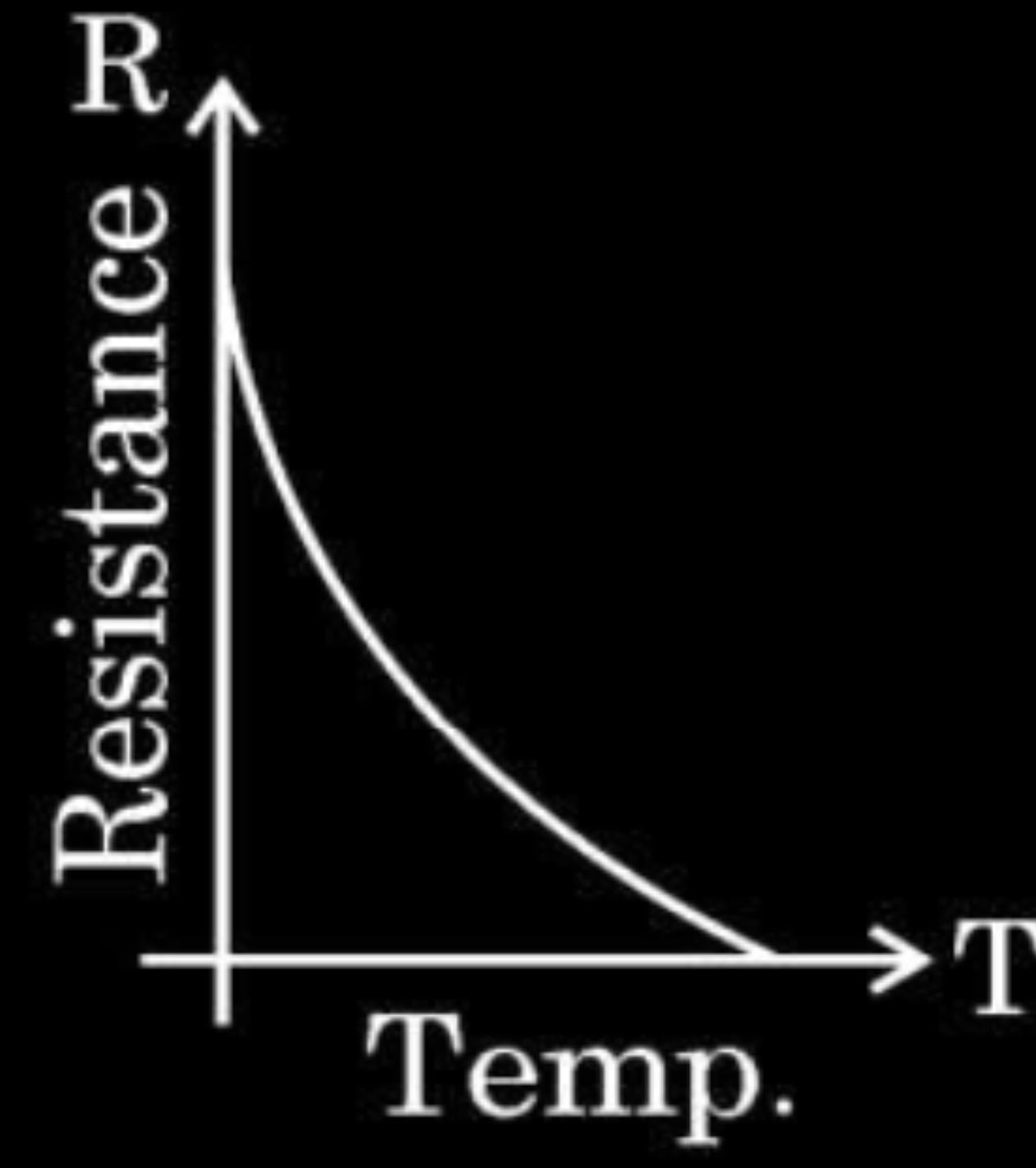
2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



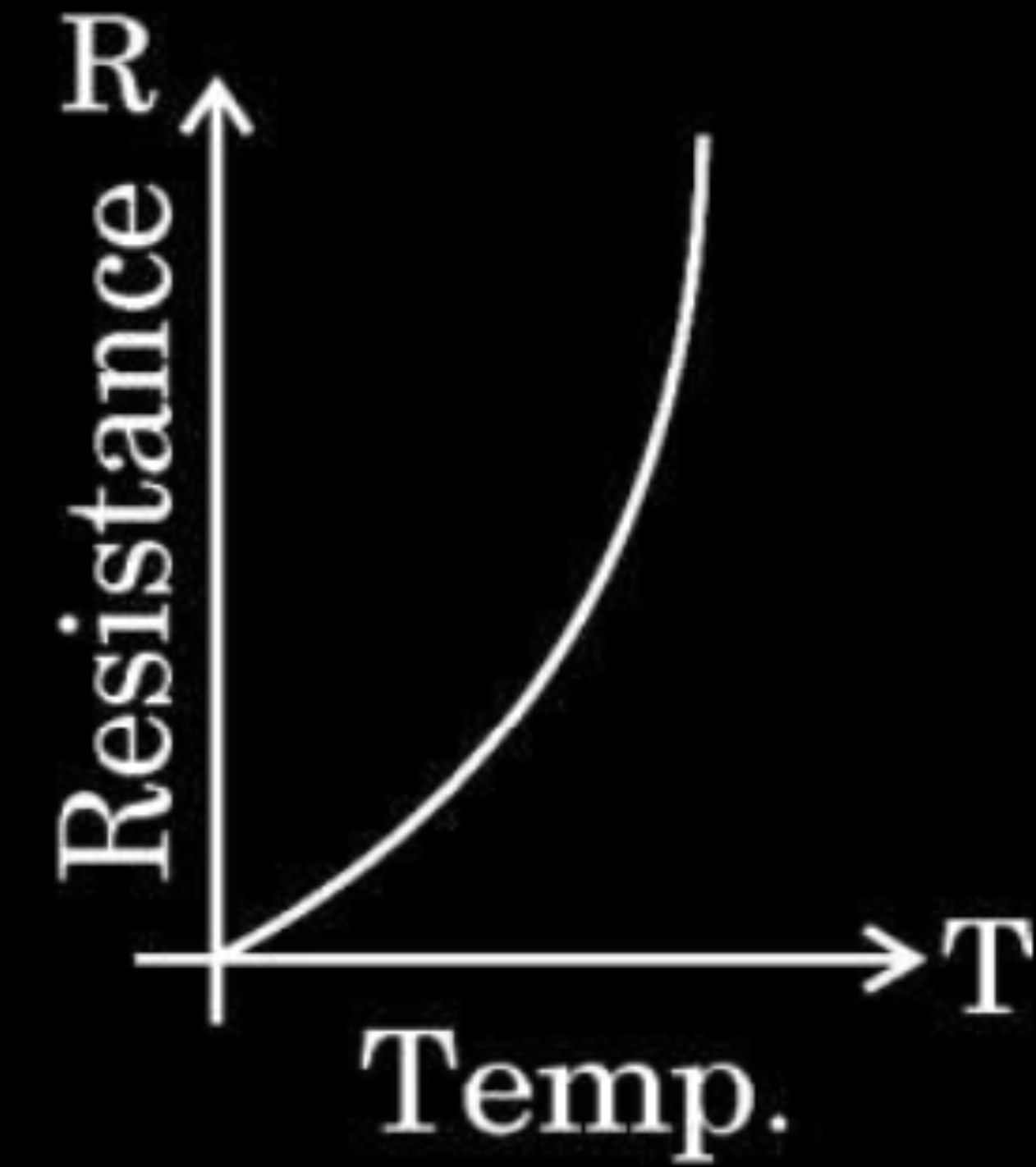
(a)



(b)

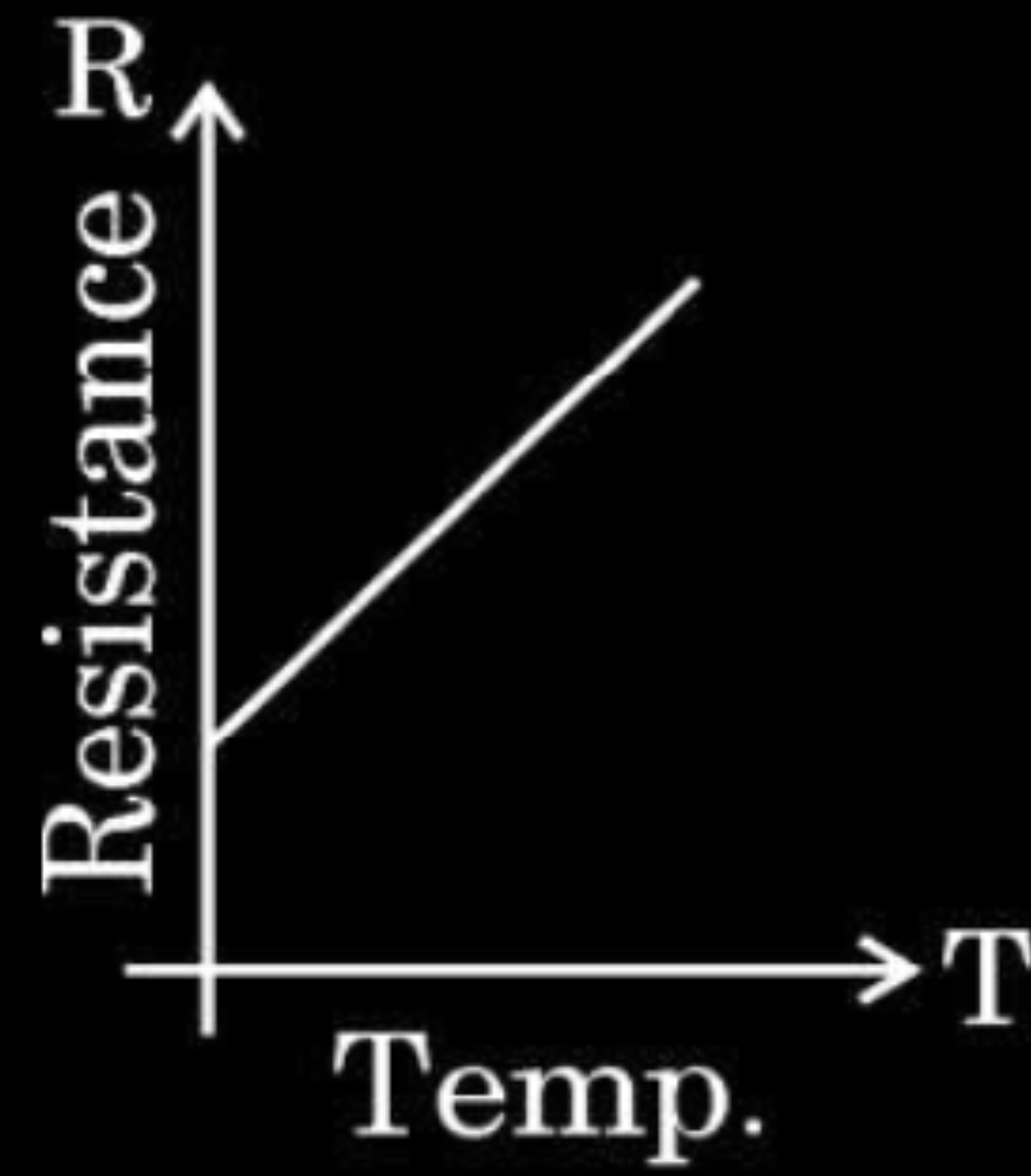


(c)

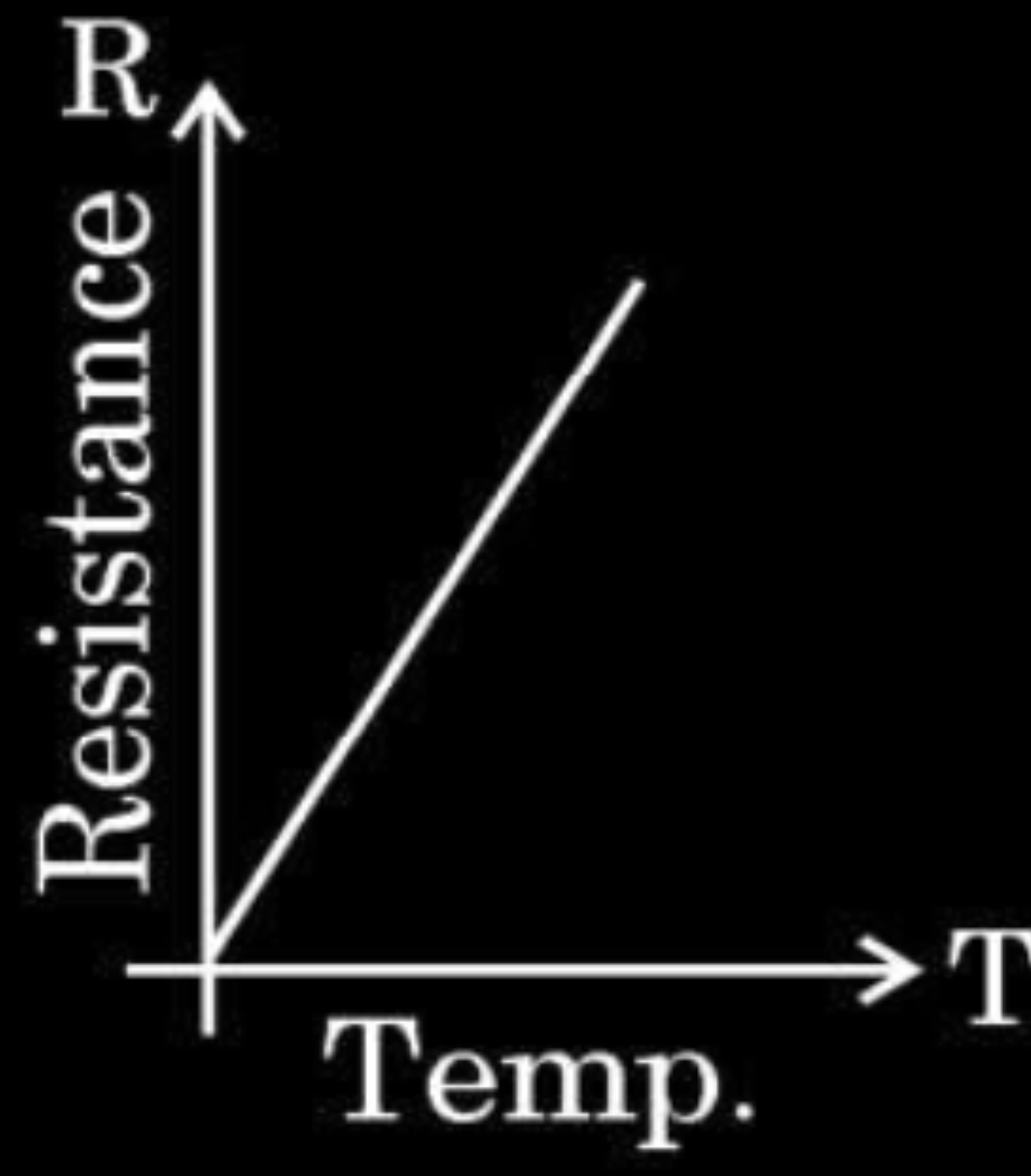


(d)

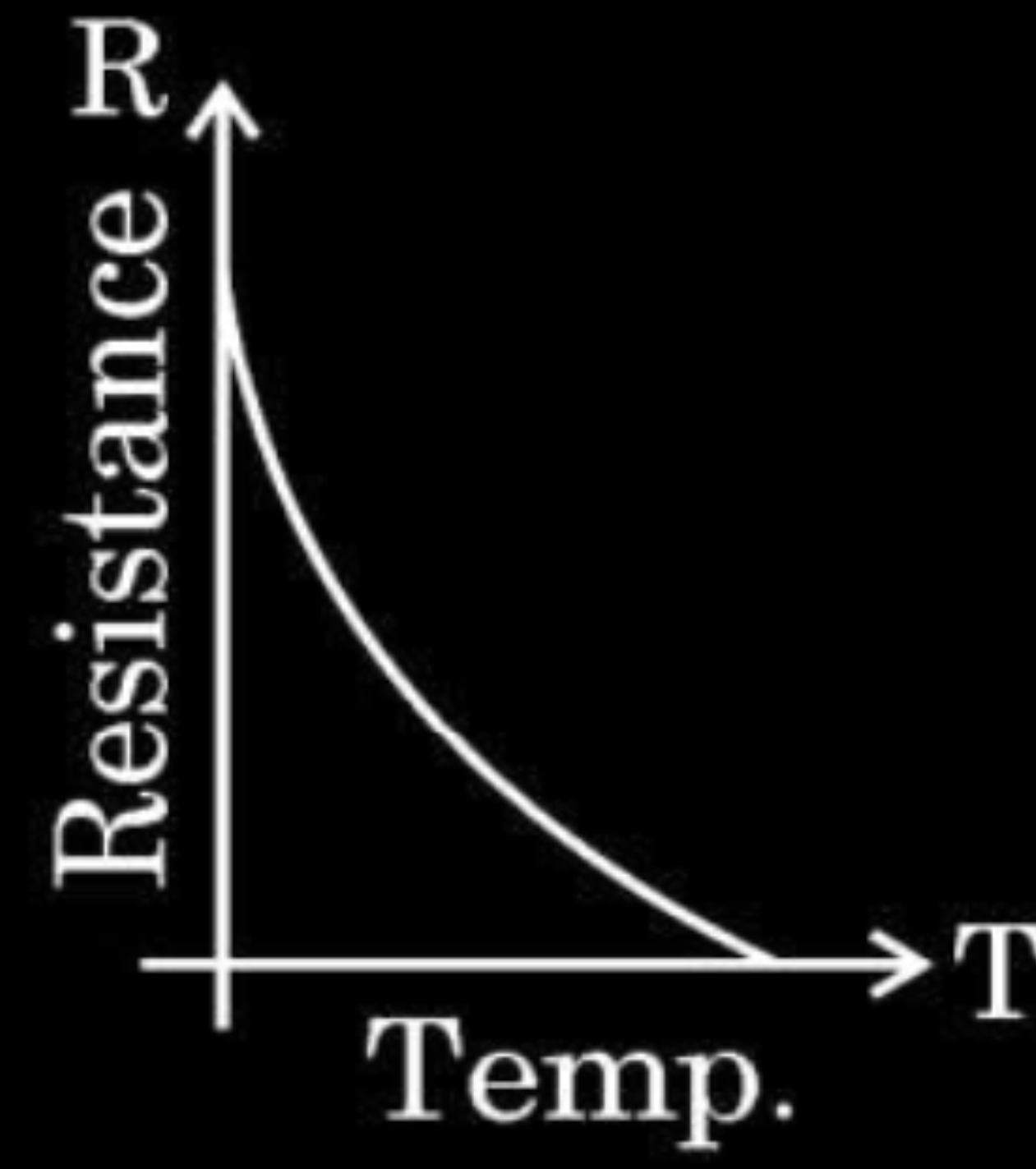
2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



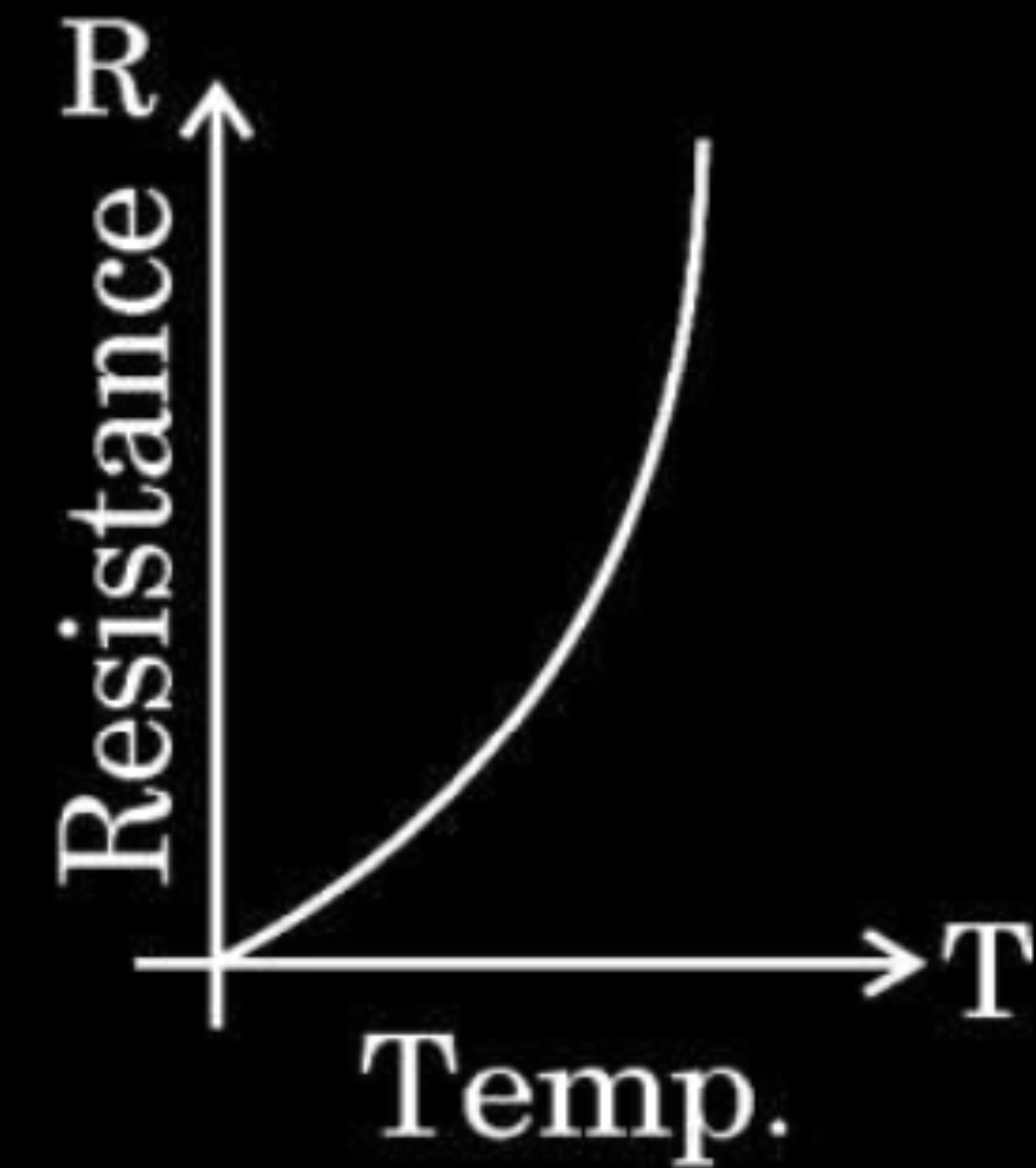
(a)



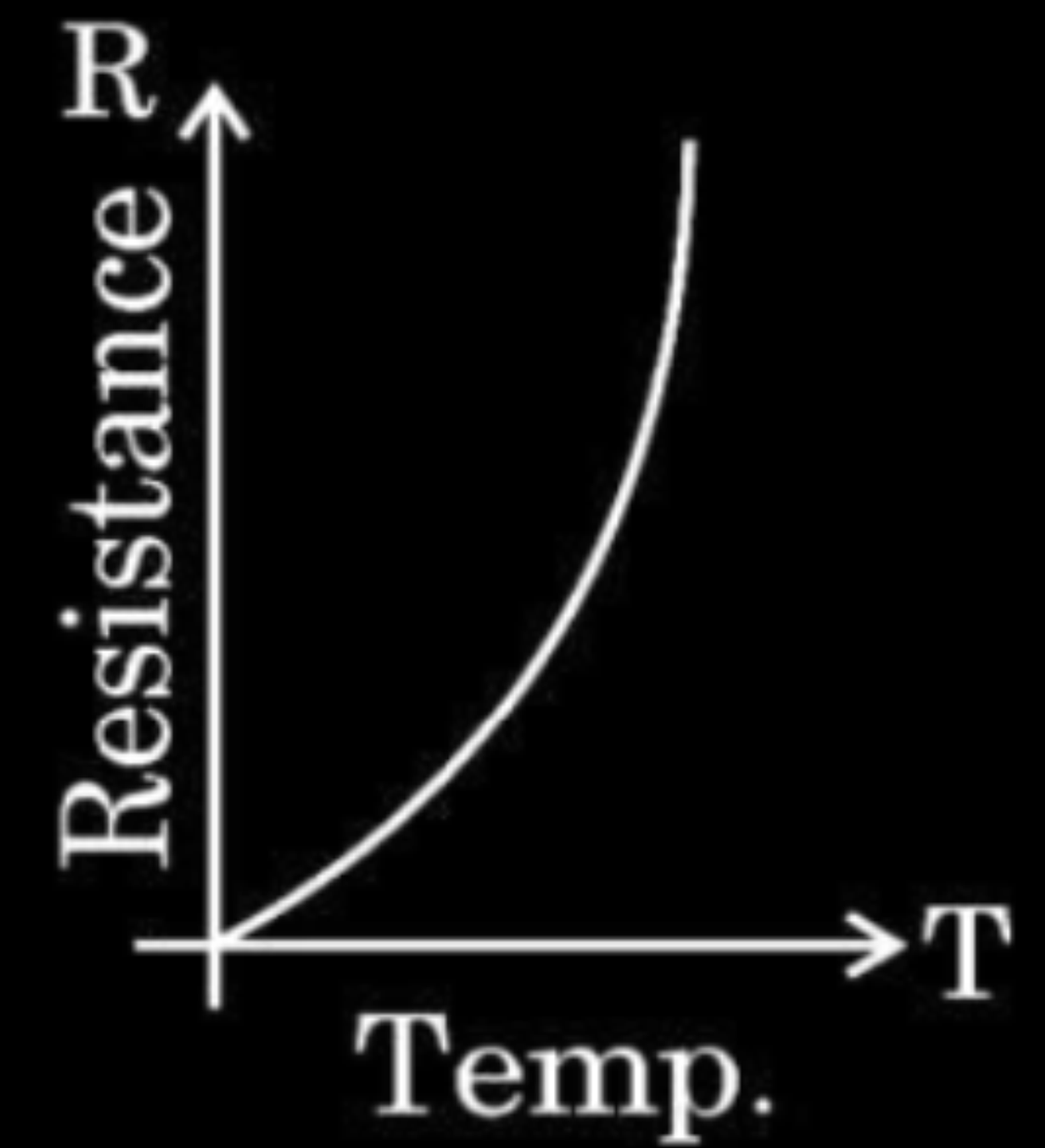
(b)



(c)



(d)



(d)

9. A conductor of $10\ \Omega$ is connected across a $6\ \text{V}$ ideal source. The power supplied by the source to the conductor is **1**
- (a) $1.8\ \text{W}$ (b) $2.4\ \text{W}$
(c) $3.6\ \text{W}$ (d) $7.2\ \text{W}$

9. A conductor of $10\ \Omega$ is connected across a $6\ \text{V}$ ideal source. The power supplied by the source to the conductor is **1**
- (a) $1.8\ \text{W}$ (b) $2.4\ \text{W}$
(c) $3.6\ \text{W}$ (d) $7.2\ \text{W}$

(c) $3.6\ \text{W}$

3. A current of 0.8 A flows in a conductor of $40\ \Omega$ for 1 minute. The heat produced in the conductor will be 1
- (a) 1445 J (b) 1536 J
- (c) 1569 J (d) 1640 J

3. A current of 0.8 A flows in a conductor of $40\ \Omega$ for 1 minute. The heat produced in the conductor will be

- (a) 1445 J (b) 1536 J
- (c) 1569 J (d) 1640 J

(b)
1536 J

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T . The conductivity of

1

- (a) both increases.
- (b) both decreases.
- (c) copper increases and silicon decreases.
- (d) copper decreases and silicon increases.

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T . The conductivity of 1
- (a) both increases.
 - (b) both decreases.
 - (c) copper increases and silicon decreases.
 - (d) copper decreases and silicon increases.

(d)

Copper decreases and silicon increases

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

(c)Assertion (A) is true, but Reason (R) is false.

- 25.** (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ?

2

OR

- (b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R , $2R$ and $3R$. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer.

2

(a) For Answer
Reason

1
1

For series Answer
Justification

$\frac{1}{2}$
 $\frac{1}{2}$

For parallel Answer
Justification

$\frac{1}{2}$
 $\frac{1}{2}$

(a) Power delivered just after it is turned on, will be more because resistance of the bulb is low. After some time temperature of the bulb increases and resistance also increases and therefore power $(\frac{V^2}{R})$ becomes low.	1 1
For series Power dissipated will be maximum for 3R. Because current is same and power is proportional to resistance.	$\frac{1}{2}$ $\frac{1}{2}$
For parallel Power dissipated will be maximum for R. Because voltage is same and power is inversely proportional to resistance.	$\frac{1}{2}$ $\frac{1}{2}$

CBSE - 2024

1. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be

(A) P

(B) $\frac{P}{2}$

(C) $\frac{P}{4}$

(D) $2P$

1. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be

(A) P

(B) $\frac{P}{2}$

(C) $\frac{P}{4}$

(D) $2P$

(D) $2P$

24. (i) Define ‘temperature coefficient of resistance’ of a metal.
- (ii) Show the variation of resistivity of copper with rise in temperature.
- (iii) The resistance of a wire is $10\ \Omega$ at $27\ ^\circ\text{C}$. Find its resistance at $-73\ ^\circ\text{C}$.
The temperature coefficient of resistance of the material of the wire is $1.70 \times 10^{-4}\ ^\circ\text{C}^{-1}$.

(i) Defining temperature coefficient	1
(ii) Showing the variation of resistivity	1
(iii) Finding the resistance	1

(i) Change in resistance per unit original resistance per degree change in temperature is temperature coefficient of resistance.

$$\begin{aligned} \text{(iii) } R_2 &= R_1 (\theta_2 - \theta_1) \alpha + R_1 \\ &= 10(-73 - 27) \times 1.70 \times 10^{-4} + 10 \\ &= -0.170 + 10 \end{aligned}$$

$$R_2 = 9.83 \Omega$$

Alternatively

$$R_1 = R_0 (1 + \alpha t_1)$$

$$R_2 = R_0 (1 + \alpha t_2)$$

$$\frac{R_1}{R_2} = \frac{(1 + \alpha t_1)}{(1 + \alpha t_2)}$$

$$R_2 = \frac{(1 + \alpha t_1)}{(1 + \alpha t_2)} R_1$$

$$R_2 = \left[\frac{1 + 1.70 \times 10^{-4} \times (-73)}{1 + 1.70 \times 10^{-4} \times 27} \right] \times 10$$

$$R_2 = \frac{0.98759}{1.00459} \times 10 \Omega$$

$$R_2 = 9.83 \Omega$$

1

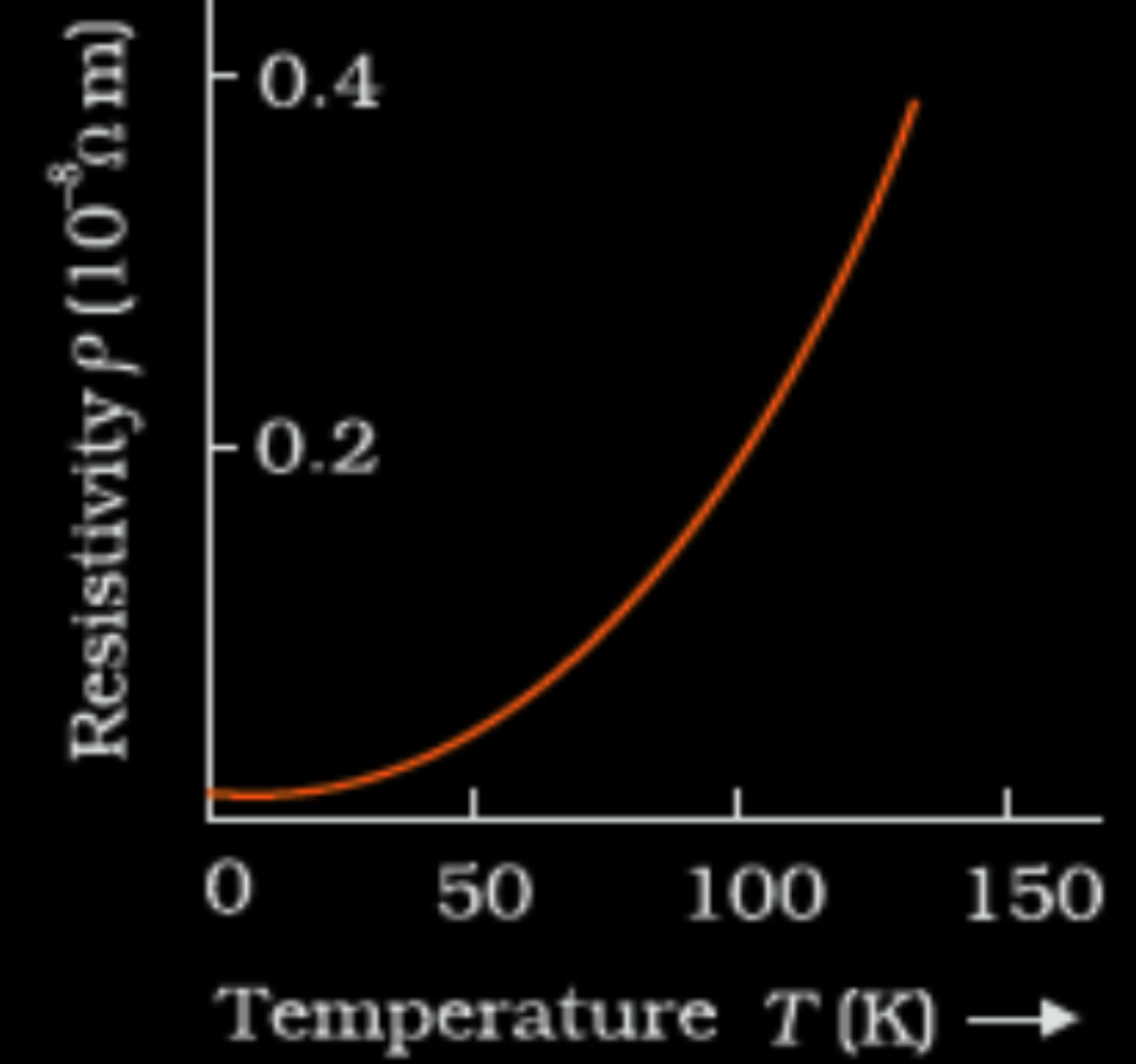
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(ii)



20. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is $2.0 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$.

20. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is $2.0 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$.

2

$R_2 = R_1 + 25\% \text{ of } R_1 = 1.25R_1$	$\frac{1}{2}$
Temperature coefficient of resistance	
$\alpha = \frac{R_2 - R_1}{R_1 \cdot \Delta T}$	$\frac{1}{2}$
$T_2 - 27 = \frac{1.25R_1 - R_1}{R_1 \times 2 \times 10^{-4}}$	$\frac{1}{2}$
$T_2 = 1277 \text{ }^{\circ}\text{C}$	$\frac{1}{2}$

20. Two electric heaters have power ratings P_1 and P_2 , at voltage V . They are connected in series to a dc source of voltage V . Find the power consumed by the combination. Will they consume the same power if connected in parallel across the same source ?

2

Finding power consumed by two electric heaters in series combination	$1 \frac{1}{2}$
Writing answer for parallel combination	$\frac{1}{2}$

$R_1 = \frac{V^2}{P_1} \quad \& \quad R_2 = \frac{V^2}{P_2}$	$\frac{1}{2}$
$R_{eq} = R_1 + R_2 = V^2 \left(\frac{1}{P_1} + \frac{1}{P_2} \right)$	$\frac{1}{2}$
$P_{series} = \frac{V^2}{R_{eq}}$	
$P_{series} = \frac{V^2}{V^2 \left(\frac{1}{P_1} + \frac{1}{P_2} \right)}$	
$\frac{1}{P_{series}} = \frac{1}{P_1} + \frac{1}{P_2}$	$\frac{1}{2}$
No	$\frac{1}{2}$

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C . Take 20°C as the reference temperature and temperature coefficient of resistance of silver at $20^{\circ}\text{C} = 4.0 \times 10^{-3} \text{ K}^{-1}$.

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C . Take 20°C as the reference temperature and temperature coefficient of resistance of silver at $20^{\circ}\text{C} = 4.0 \times 10^{-3} \text{ K}^{-1}$.

2

Finding the temperature	2
$R = R_0 [1 + \alpha (T - T_0)]$	$\frac{1}{2}$
$R = 2 R_0$ [Given]	
$2 R_0 = R_0 [1 + \alpha (T - T_0)]$	$\frac{1}{2}$
On solving	
$T = T_0 + 250$	
$T = 270^{\circ}\text{C}$ or 543 K	1

22. (a) Define the terms – ‘temperature coefficient of resistance’ and ‘electrical conductivity’ of a conductor. Why are constantan and manganin used for making standard resistances ? Explain.

3

Definition of (i) Temperature coefficient of resistance	1
(ii) Electrical Conductivity	1
Explanation of choosing constantan and manganin for making standard resistances.	1

Temperature coefficient of resistance is defined as the fractional change in resistance per unit increase in temperature.	1
Electrical conductivity is defined as current density per unit electric field.	1
Alternatively: Electrical conductivity is the reciprocal of electrical resistivity.	
<u>Reason:</u> Alloys have high resistivity Low temperature coefficient of resistance Ability to achieve more resistance value in small size (Any Two)	$\frac{1}{2} + \frac{1}{2}$

- 3.** A steady current I is passed through a conductor at room temperature for time t . It is observed that its temperature rises by 0.5°C . If $2I$ current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately :

(A) 1.0°C

(B) 1.5°C

(C) 2.0°C

(D) 4.0°C

3. A steady current I is passed through a conductor at room temperature for time t . It is observed that its temperature rises by 0.5°C . If $2I$ current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately :

(A) 1.0°C

(B) 1.5°C

(C) 2.0°C

(D) 4.0°C

(C) 2.0°C

(ii) The temperature coefficient of resistance of nichrome is $1.70 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$. In order to increase resistance of a nichrome wire by 8.5%, the temperature of the wire should be increased by : 1

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

(ii) The temperature coefficient of resistance of nichrome is $1.70 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$. In order to increase resistance of a nichrome wire by 8.5%, the temperature of the wire should be increased by :

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

1. A wire of resistance R , connected to an ideal battery consumes a power P . If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

(A) $\frac{P}{4}$

(B) $\frac{P}{2}$

(C) P

(D) $2P$

1. A wire of resistance R , connected to an ideal battery consumes a power P . If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

(A) $\frac{P}{4}$

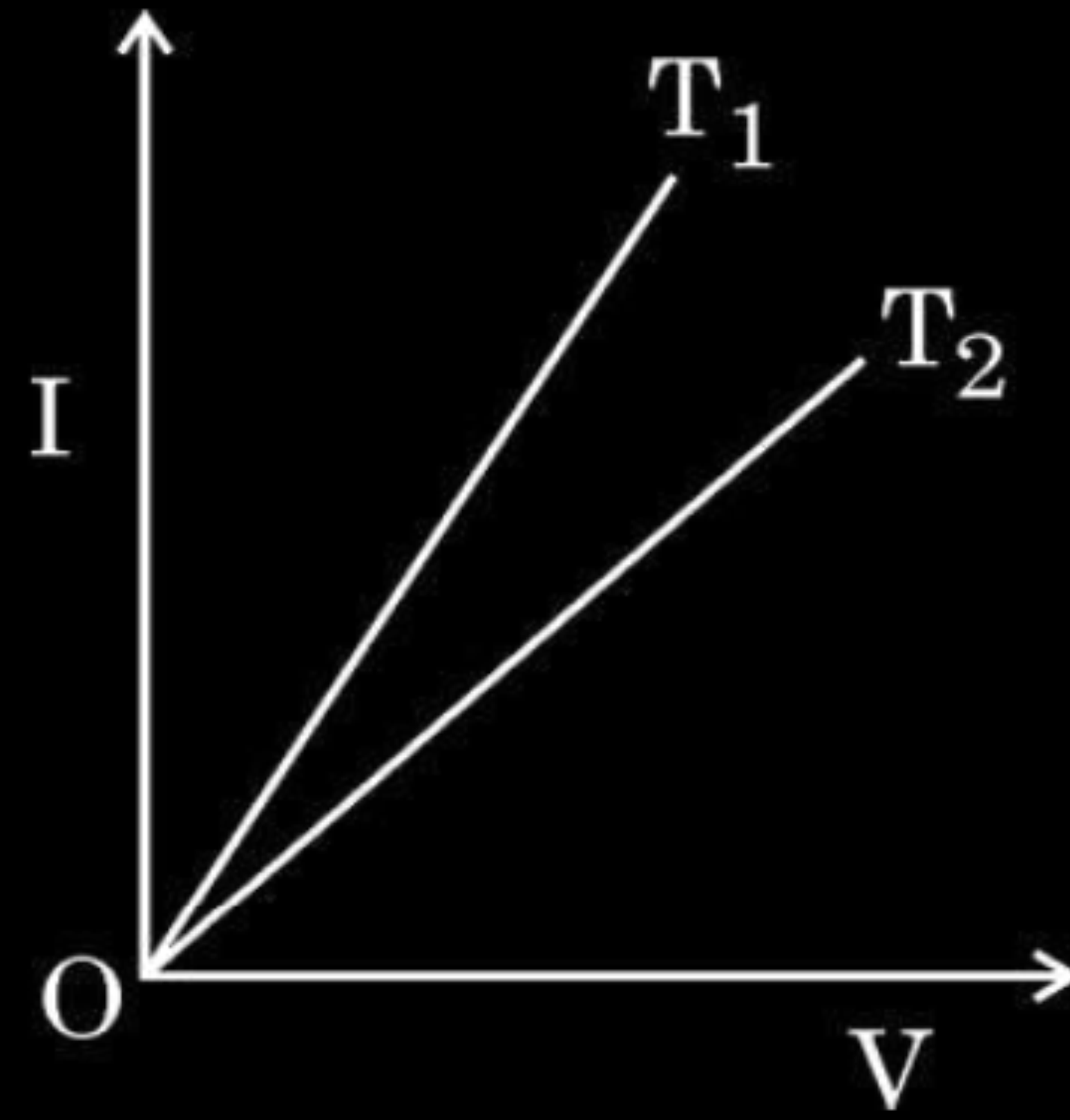
(B) $\frac{P}{2}$

(C) P

(D) $2P$

(A) $\frac{P}{4}$

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2 . One can conclude that :



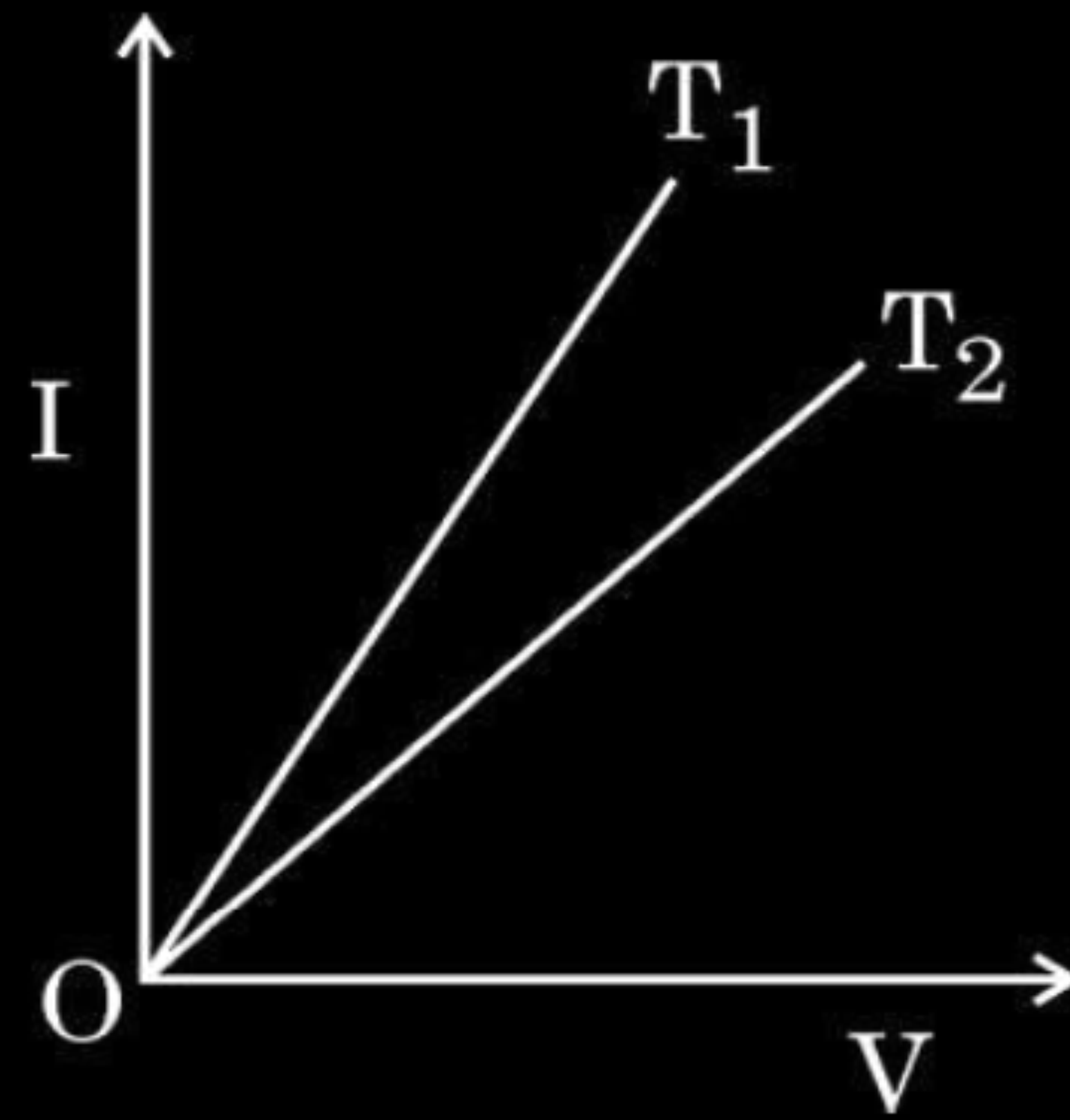
(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $T_1 < T_2$

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2 . One can conclude that :



(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $T_1 < T_2$

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R , in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2 - 1}{n^2}\right)R$

(B) $\left(\frac{n^2 + 1}{n^2 - 1}\right)R$

(C) $\left(\frac{n^2 - 1}{n}\right)R$

(D) $\frac{(n^2 + 1)R}{n^2}$

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R , in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2 - 1}{n^2}\right)R$

(B) $\left(\frac{n^2 + 1}{n^2 - 1}\right)R$

(C) $\left(\frac{n^2 - 1}{n}\right)R$

(D) $\frac{(n^2 + 1)R}{n^2}$

2. The resistance of a wire at $20\text{ }^{\circ}\text{C}$ is $50\ \Omega$. When it is heated to $120\text{ }^{\circ}\text{C}$, the resistance becomes $51\ \Omega$. The temperature coefficient of resistance of the material of the wire is **1**

(A) $2 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(B) $1 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(C) $5 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(D) $1 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$

2. The resistance of a wire at $20\text{ }^{\circ}\text{C}$ is $50\ \Omega$. When it is heated to $120\text{ }^{\circ}\text{C}$, the resistance becomes $51\ \Omega$. The temperature coefficient of resistance of the material of the wire is

1

(A) $2 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(B) $1 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(C) $5 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

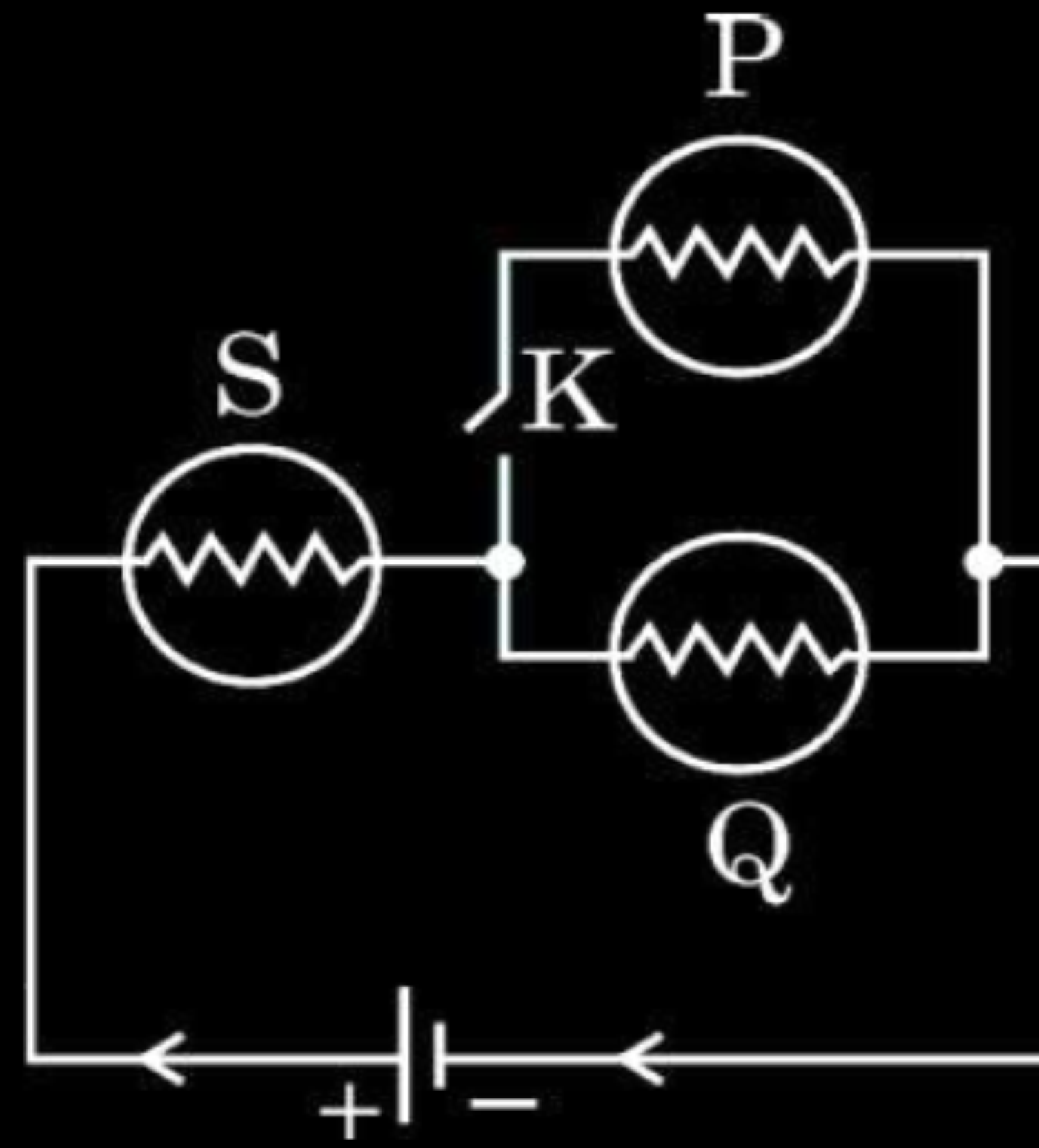
(D) $1 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$

- 21.** The resistance of a wire at 25°C is $10.0\ \Omega$. When heated to 125°C , its resistance becomes $10.5\ \Omega$. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C . 2

- 21.** The resistance of a wire at 25°C is $10.0\ \Omega$. When heated to 125°C , its resistance becomes $10.5\ \Omega$. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C . 2

Finding (i) temperature coefficient of resistance 1 1/2 (ii) resistance of wire at 425°C 1/2	
(i) $R_2 = R_1(1 + \alpha(t_2 - t_1))$ $10.5 = 10(1 + \alpha \times 100)$	1/2
$\alpha = 5 \times 10^{-4} / ^{\circ}\text{C}$	1/2
(ii) $R_{425} = R_{25}(1 + \alpha(425 - 25))$ $= 10(1 + 5 \times 10^{-4} \times 400)$	1/2
$= 12\ \Omega$	1/2

21. (a) In the given figure, three identical bulbs P, Q and S are connected to a battery.



- (i) Compare the brightness of bulbs P and Q with that of bulb S when key K is closed.
- (ii) Compare the brightness of the bulbs S and Q when the key K is opened.

Justify your answer in both cases.

2

(i) Comparison of brightness of bulbs P and Q with bulb S	$\frac{1}{2}$
Justification	$\frac{1}{2}$
(ii) Comparison of brightness of bulb S with Q	$\frac{1}{2}$
Justification	$\frac{1}{2}$

(i) Brightness of the bulb 'S' will be more than bulbs 'P' and 'Q' The current flowing through the bulb 'S' is twice of the current in bulbs 'P' and 'Q'.	$\frac{1}{2}$ $\frac{1}{2}$
(ii) Brightness of the bulb 'S' and 'Q' will be same The current flowing through both bulbs is same.	$\frac{1}{2}$ $\frac{1}{2}$
Alternatively-	
(i) Brightness of the bulb 'S' will be more than bulbs 'P' and 'Q' The potential difference across 'S' is twice than the potential difference across bulbs 'P' and 'Q'	
(ii) Brightness of both bulbs 'S' and 'Q' is same. The potential difference across 'S' and 'Q' will be same.	

6. The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, then the power consumed by it will be

1

- (A) P
- (B) 2P
- (C) $\frac{P}{2}$
- (D) $\frac{P}{4}$

6. The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, then the power consumed by it will be

1

(A) P

(B) $2P$

(C) $\frac{P}{2}$

(D) $\frac{P}{4}$

4. Two resistors R_1 and R_2 of $4\ \Omega$ and $6\ \Omega$ are connected in parallel across a battery. The ratio of power dissipated in them, $P_1 : P_2$ will be *1*

(A) $4 : 9$

(B) $3 : 2$

(C) $9 : 4$

(D) $2 : 3$

4. Two resistors R_1 and R_2 of $4\ \Omega$ and $6\ \Omega$ are connected in parallel across a battery. The ratio of power dissipated in them, $P_1 : P_2$ will be 1

(A) $4 : 9$

(B) $3 : 2$

(C) $9 : 4$

(D) $2 : 3$

- (a) Draw a graph showing the variation of current versus voltage in an electrolyte when an external resistance is also connected.
- (b) (i) The graph between resistance (R) and temperature (T) for Hg is shown in the figure (a). Explain the behaviour of Hg near 4 K.

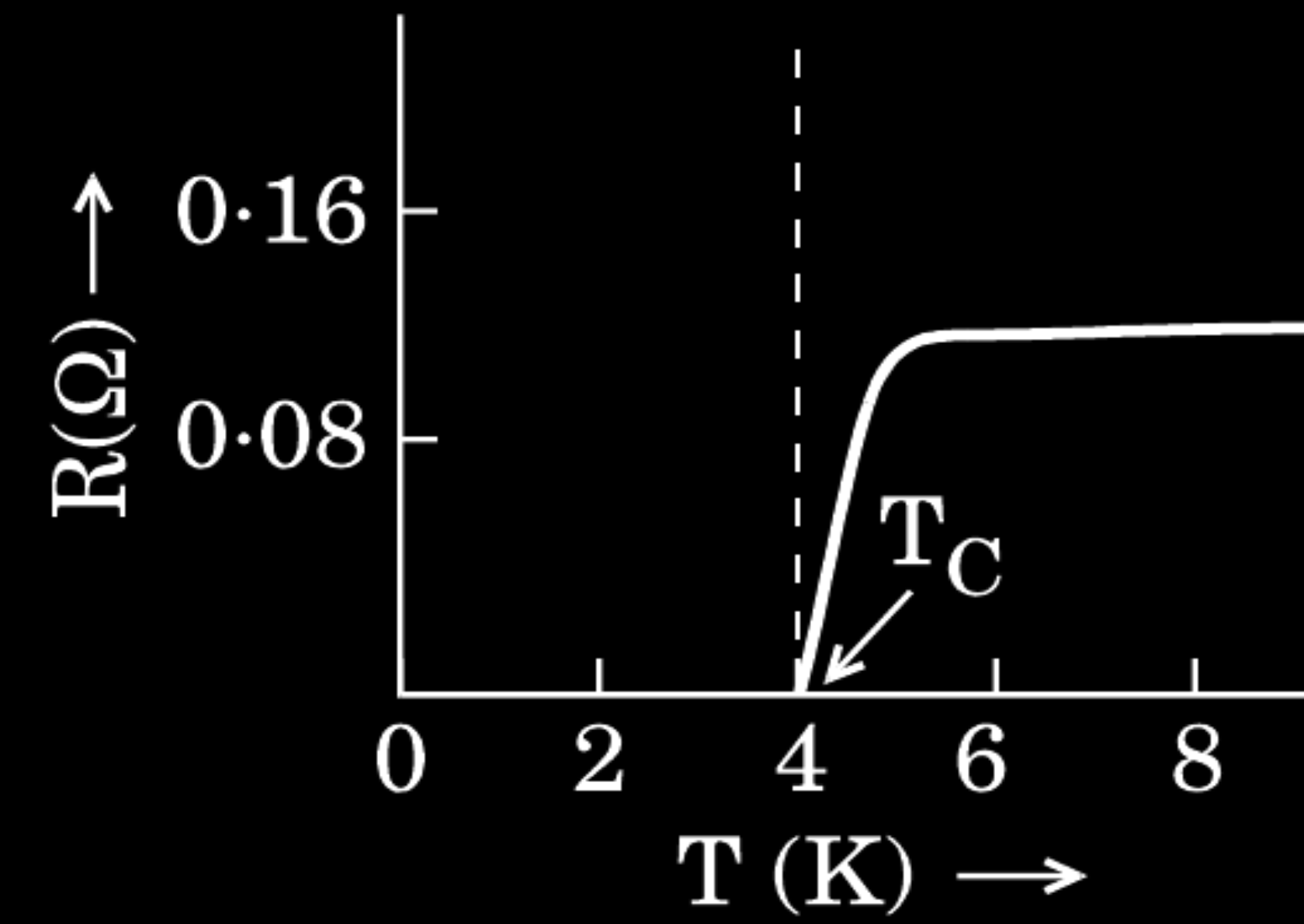


Figure (a)

- (ii) In which region of the graph shown in the figure (b) is the resistance negative and why?

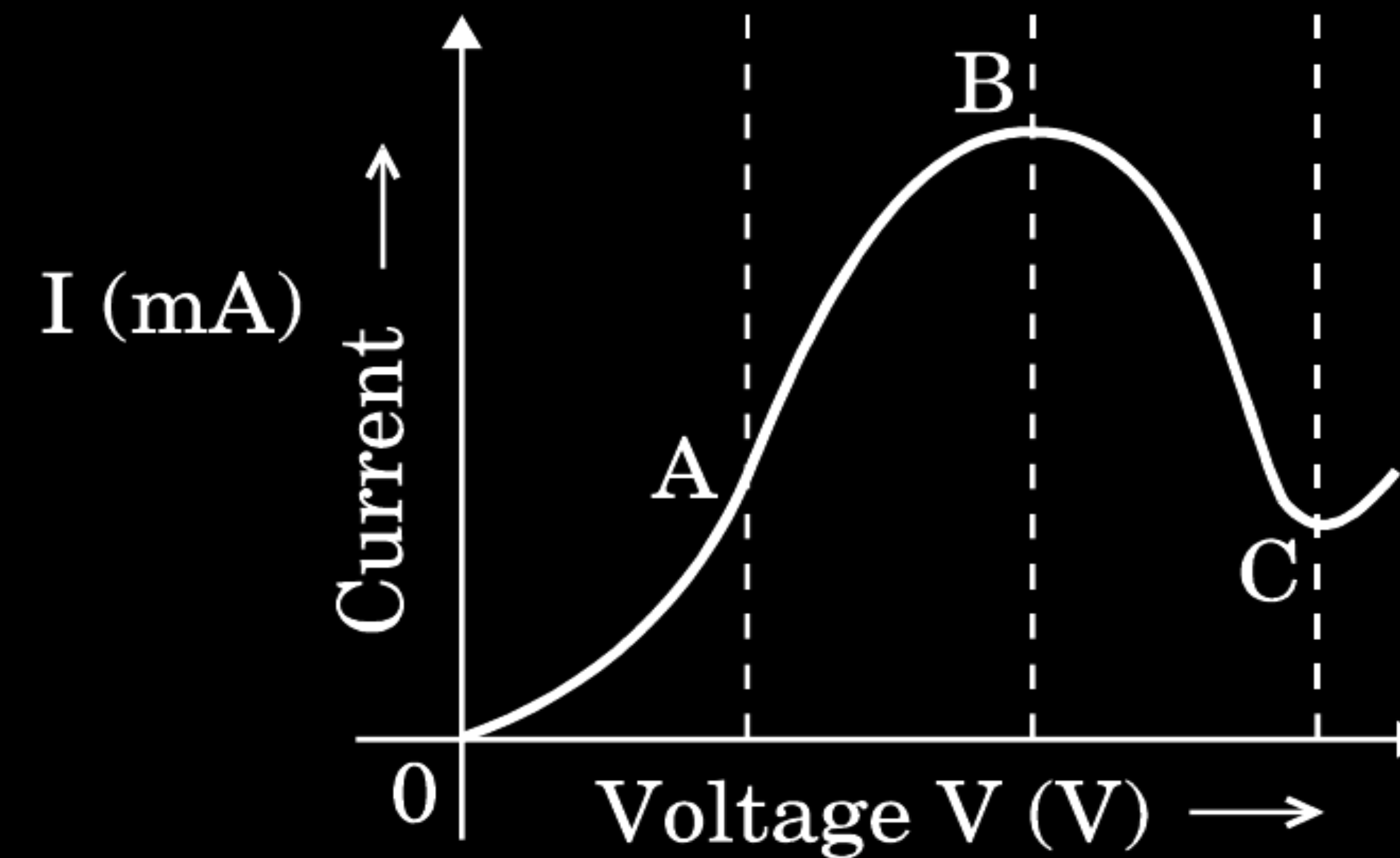
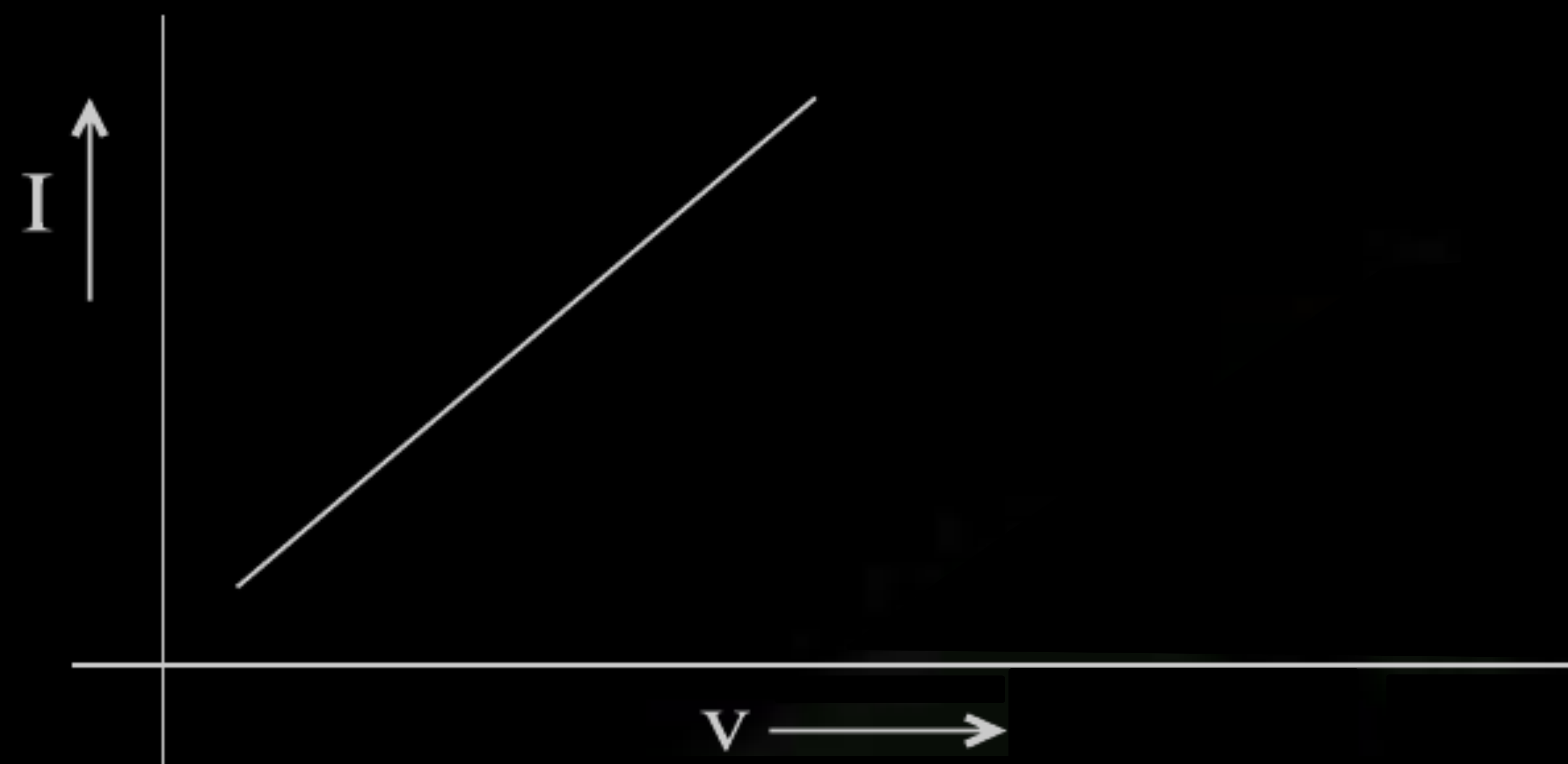
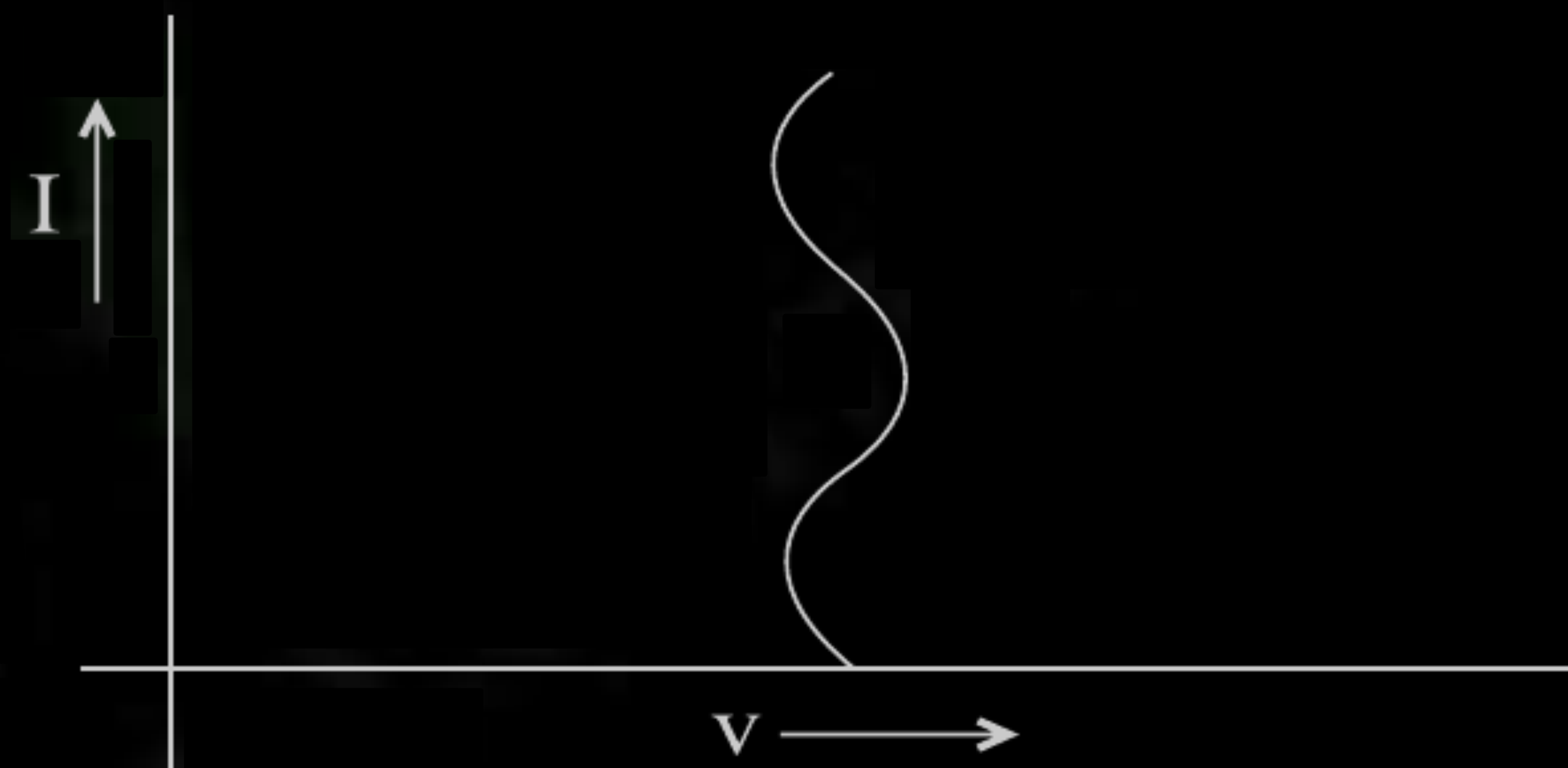


Figure (b)

a) Note : Award this 1 mark even if the student draw the following or some other non linear graph



OR



1

- b) At a temperature of 4 K, the Resistance of Hg becomes zero.
Note - Award this 1 mark if the student writes that Hg becomes a super conductor at temperature of 4 K
- c) Region BC ; since current is decreasing with increasing voltage
Alternately : The graph has negative slope for this region.

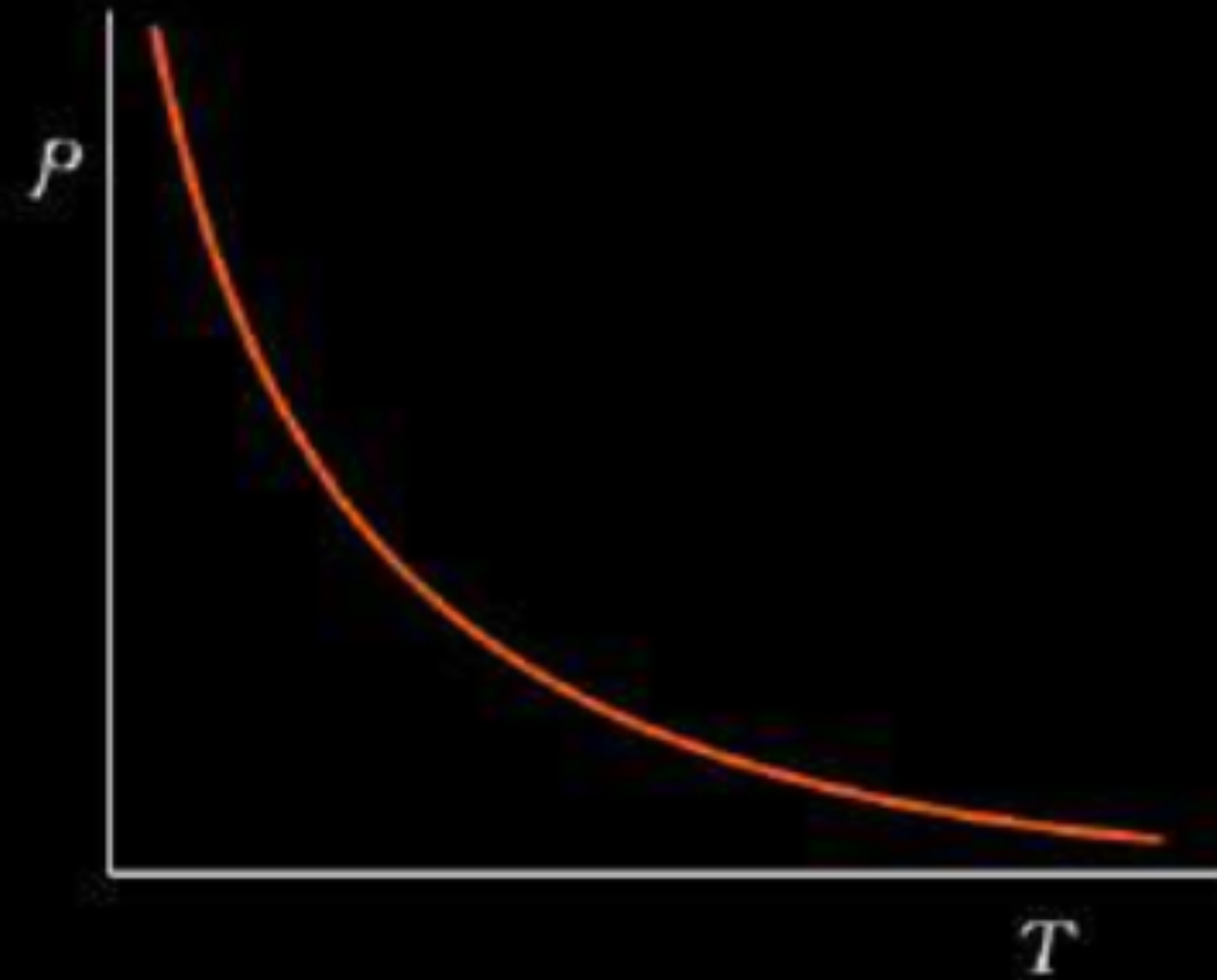
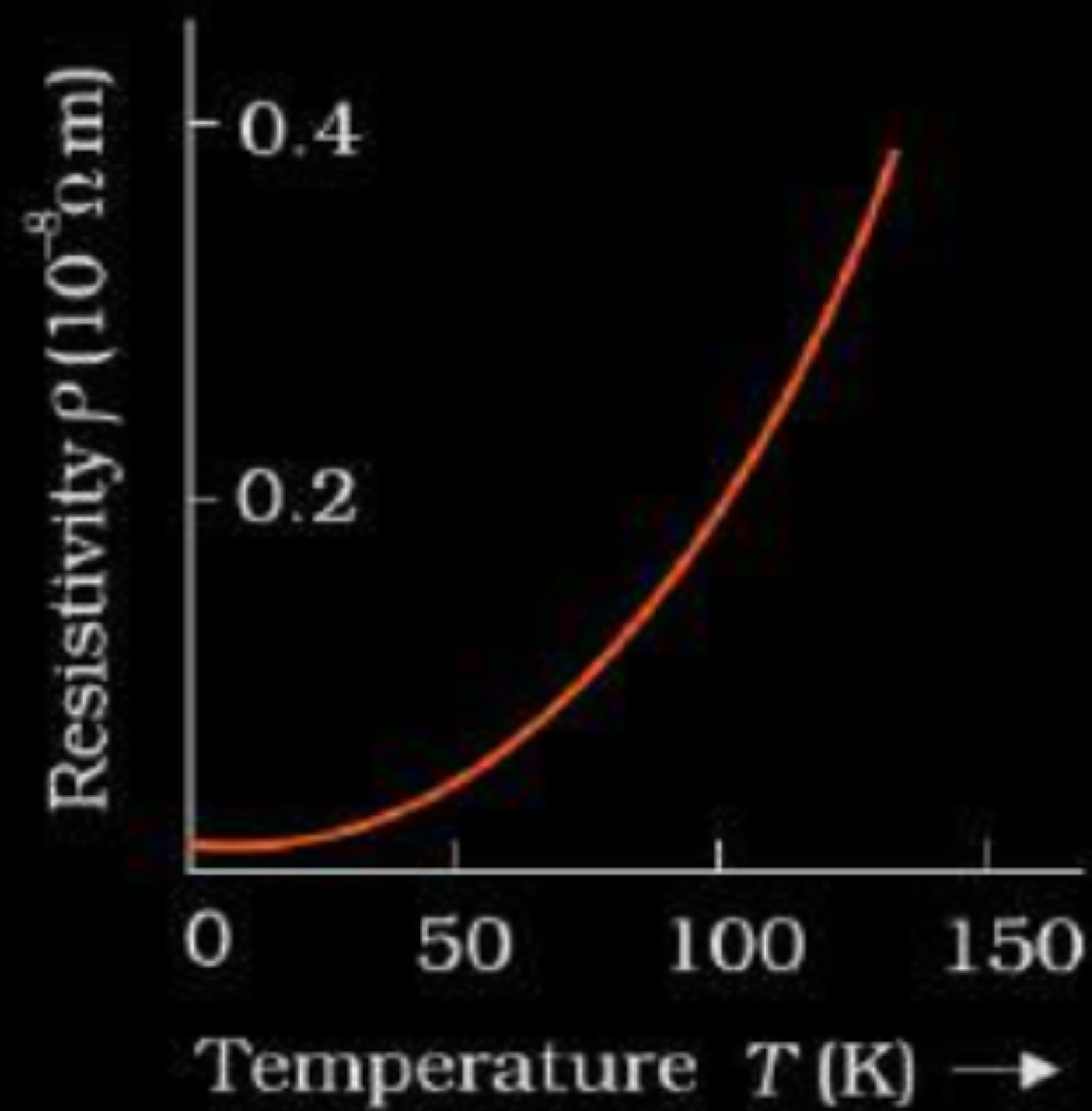
1

1

Show, on a plot, variation of resistivity of (i) a conductor, and (ii) a typical semiconductor as a function of temperature.

Using the expression for the resistivity in terms of number density and relaxation time between the collisions, explain how resistivity in the case of a conductor increases while it decreases in a semiconductor, with the rise of temperature.

- | | |
|---|-----------------------------|
| • Showing the plot of variation of resistivity | $\frac{1}{2} + \frac{1}{2}$ |
| • Expression for resistivity | 1 |
| • Explaining variation of resistivity for conductor and semiconductor | $\frac{1}{2} + \frac{1}{2}$ |



- $\rho = \frac{m}{ne^2\tau}$
- In case of conductors with increase in temperature, relaxation time decreases, so resistivity increases.
- In case of semiconductors with increase in temperature number density (n) of free electrons increases, hence resistivity increases.

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

Two bulbs are rated (P_1, V) and (P_2, V) . If they are connected (i) in series and (ii) in parallel across a supply V , find the power dissipated in the two combinations in terms of P_1 and P_2 .

Two bulbs are rated (P_1, V) and (P_2, V) . If they are connected (i) in series and (ii) in parallel across a supply V , find the power dissipated in the two combinations in terms of P_1 and P_2 .

$$R_1 = \frac{V^2}{P_1} \quad , \quad R_2 = \frac{V^2}{P_2} \quad ,$$

$$P_s = \frac{V^2}{R_s} = \frac{P_1 P_2}{P_1 + P_2}$$

$$\frac{1}{P_s} = \frac{1}{P_1} + \frac{1}{P_2}$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{P_1 + P_2}{V^2}$$

$$\therefore P_p = \frac{V^2}{R_p} = P_1 + P_2$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Two electric bulbs P and Q have their resistances in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs.

Two electric bulbs P and Q have their resistances in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs.

Power = $I^2 R$	
The current, in the two bulbs, is the same as they are connected in series.	$\frac{1}{2}$
$\therefore \frac{P_1}{P_2} = \frac{I^2 R_1}{I^2 R_2} = \frac{R_1}{R_2}$	$\frac{1}{2}$
$= \frac{1}{2}$	$\frac{1}{2}$

Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more ? Justify your answer.

Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more ? Justify your answer.

i. Nichrome	$\frac{1}{2}$
ii. $R_{\text{Ni}} > R_{\text{Cu}}$ (or Resistivity _{Ni} > Resistivity _{Cu})	$\frac{1}{2}$

- (a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change ?

- (a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change ?

a) $H = \frac{V^2}{R}$

$\therefore V$ increases by a factor of $\sqrt{9} = 3$

$\frac{1}{2}$

$\frac{1}{2}$